

Pixel IT Center

Website Modernization & Portal Foundation

Recommended architecture, technology stack, website structure, rollout plan, and foundation for portal.pixelitcenter.com

PHASE 0	PHASE 1	FOUNDATION
Modernize public website	Launch operations portal	One scalable technology direction

Decision in one sentence

Keep the current public site live while the team builds a new Next.js/TypeScript website in staging; launch it first, then build the secure operations portal as a separate application using the same design system.

Executive Summary

Pixel IT Center should modernize the public website first and treat the internal operations portal as a separate product. This reduces risk, gives the company a professional public presence quickly, and establishes a reusable design and engineering foundation for the portal.

Recommended direction

Build the new public website at staging.pixelitcenter.com using Next.js + TypeScript + Tailwind CSS. Keep pixelitcenter.com live during development. After business approval, switch the production domain to the new site. Build portal.pixelitcenter.com separately afterward.

Why this sequence works

- **Lower risk.** The current website remains available while the replacement is built and tested.
- **Cleaner architecture.** Marketing pages and internal business workflows do not become one oversized application.
- **Reusable design system.** The visual language created for the public site can be reused in the portal.
- **Faster delivery.** The public website does not need the full database, authentication, and workflow stack required by the portal.
- **Better long-term ownership.** Both applications can evolve independently while sharing branding and engineering standards.

Current site observations

The current Pixel IT Center website already contains the core content needed for a modern rebuild: company overview, mission and vision, service areas, client references, testimonials, and contact information. The main opportunity is to improve information architecture, copy quality, visual hierarchy, mobile usability, calls to action, performance, and maintainability.

Source reviewed: <https://pixelitcenter.com> on September 27, 2026.

Target Architecture

Keep the public website and internal portal separate

pixelitcenter.com - PUBLIC WEBSITE
Home Services Staffing About Careers Contact
Next.js + TypeScript + Tailwind CSS
Public content, lead generation, recruiting, brand credibility

portal.pixelitcenter.com - INTERNAL OPERATIONS PORTAL
Clients Employees Assignments Timesheets Invoices Payments Reports
Next.js + NestJS + PostgreSQL + Cognito
Secure business operations and source of truth

Deployment approach

CURRENT: pixelitcenter.com stays live
BUILD: staging.pixelitcenter.com
REVIEW: design, content, mobile, forms, SEO, performance
LAUNCH: point pixelitcenter.com to the new website
NEXT: begin portal.pixelitcenter.com
Important boundary Do not place timesheets, invoices, employee records, or other internal workflows inside the public website codebase. The public website can link to the portal login, but the portal should remain separately secured.

Recommended Technology Stack

Public Website - Phase 0	Internal Portal - Phase 1
• Next.js • TypeScript • Tailwind CSS • shadcn/ui where useful • AWS Amplify or equivalent hosting • Google Analytics / Search Console • Gmail for contact notifications • Terraform for infrastructure where practical	• Next.js frontend • NestJS backend • TypeScript across the main stack • PostgreSQL on AWS RDS • AWS Cognito authentication • Google Drive for files initially • Gmail for notifications • AWS S3 later for document storage • Terraform for AWS infrastructure

Why TypeScript / JavaScript for the main application

Using TypeScript on both the frontend and backend gives the team one primary language, shared types, consistent tooling, and easier maintenance. This is a strong fit for client, employee, timesheet, invoice, payment, permissions, and reporting workflows.

Where Python fits

Python remains a good option for specialized services later. Use FastAPI-based Python services when there is a clear advantage, such as AI, OCR, resume parsing, document processing, analytics, forecasting, or automation. Python does not need to be the primary Phase 1 backend.

Next.js / TypeScript
NestJS / TypeScript
PostgreSQL
+ optional Python / FastAPI services for AI, data, OCR, automation

New Public Website Structure

Recommended navigation

PIXEL IT CENTER
Home Services Staffing About Us Careers Contact
Portal Login

Services

Technology Services • AI & Data • Cloud - AWS / Azure / GCP • Cybersecurity • DevOps • QA Automation • Data & Analytics • Networking / Infrastructure	Talent & Delivery • IT Staffing • Contract Staffing • Professional Services • Project / Program Support • Business Analysis • Specialized Technology Talent
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Suggested homepage flow

- 1. Clear hero statement with one primary call to action
- 2. Trusted-by / client credibility section
- 3. Core service cards
- 4. Staffing and consulting value proposition
- 5. Why Pixel IT Center
- 6. Simple delivery process
- 7. Company overview
- 8. Selected testimonials / proof points
- 9. Strong contact call to action
- 10. Footer with services, company links, contact details, privacy and terms

Copy direction

Rewrite all public-facing copy for clarity, consistency, and professional tone. Keep claims specific and supportable. Use shorter sections, clearer service descriptions, stronger calls to action, and consistent terminology across the site.

UI / UX Design Direction

The new website should feel like a modern technology and staffing company: clean, credible, fast, and easy to navigate. Avoid excessive animation or decorative complexity that slows down the site or distracts from the services.

Design principles

- Strong visual hierarchy
- Generous spacing
- Clear typography
- Consistent cards and buttons
- Accessible contrast
- Mobile-first behavior
- Professional technology imagery
- Simple, purposeful motion

Reusable design system

- Brand colors
- Typography scale
- Button variants
- Form controls
- Cards
- Navigation
- Section spacing
- Icons
- Alerts / notices
- Responsive breakpoints

Homepage wireframe

HEADER + PORTAL LOGIN
HERO: Technology & Talent Built for What's Next
CLIENT / PARTNER PROOF
SERVICES GRID
STAFFING + CONSULTING
WHY PIXEL IT CENTER
DELIVERY PROCESS
ABOUT / CREDIBILITY
TESTIMONIALS
CONTACT CTA
FOOTER
Accessibility baseline Use semantic HTML, keyboard-accessible navigation and forms, visible focus states, meaningful alt text, accessible validation, sufficient contrast, and responsive tap targets.

Development & Repository Structure

Separate repositories

pixel-website/
Deploys to pixelitcenter.com
Public marketing, careers, contact, SEO

pixel-portal/
Deploys to portal.pixelitcenter.com
Secure operations application

Suggested public website project structure

Keep the website project intentionally simple:

app/ - pages and routes
components/ - reusable UI
content/ - structured website copy where appropriate
lib/ - helpers, analytics, forms, integrations
public/ - approved images and brand assets
styles/ - global design tokens
infrastructure/ - Terraform / deployment configuration
docs/ - decisions and implementation notes

Branch and release model

- main - production-ready code
- develop or short-lived feature branches - active development
- Pull requests required for material changes
- Automated checks for linting, type checking, build, and tests
- Preview deployment for review before merge
- Staging approval before production cutover

Implementation Sequence

Phase	Focus	Main Output	Gate
0A	Discovery & content inventory	Approved sitemap, content map, asset list	Scope approved
0B	Design system & wireframes	Typography, colors, components, page wireframes	Design approved
0C	Next.js build	Responsive pages and reusable components	Feature complete
0D	Content & forms	Final copy, contact flow, careers content	Business approved
0E	Quality & SEO	Mobile QA, accessibility, metadata, redirects, analytics	Launch ready
0F	Production launch	Domain cutover and monitoring	Live
1	Operations portal	Clients, employees, assignments, timesheets, invoices	Portal MVP

Detailed Phase 0 checklist

- Back up the current website and document current URLs
- Collect logo files, approved brand assets, client logos, testimonials, and service content
- Create the new repository and CI/CD pipeline
- Create the design system before building individual pages
- Build the homepage and global navigation first
- Build Services, Staffing, About, Careers, and Contact pages
- Implement contact form delivery and spam protection
- Review and approve copy
- Add SEO metadata, sitemap, robots rules, structured data where appropriate, and redirects
- Test desktop, tablet, and mobile
- Run accessibility and performance checks
- Deploy to staging and complete stakeholder review
- Cut over the production domain only after sign-off
- Retain a recoverable backup of the existing site during the transition

Content, SEO, Performance & Security

Content migration

Do not copy every current page or paragraph word-for-word. First identify which content is still accurate and valuable, then rewrite it into the new structure. Preserve high-value URLs where possible; otherwise create redirects so search engines and existing links are not broken.

SEO essentials	Performance & security
• Unique page titles and descriptions • Logical H1-H3 structure • XML sitemap • robots.txt • Canonical URLs • Redirect map • Open Graph / social metadata • Organization / service structured data where appropriate • Google Search Console • Analytics events for key CTAs	• Optimized images • Lazy loading where appropriate • Minimal client-side JavaScript • HTTPS only • Secure headers • Validated server-side forms • Rate limiting / anti-spam • Secrets outside source code • Dependency updates • Monitoring and error tracking

Core events to measure

- Contact form submitted
- Phone / email CTA clicked
- Service page viewed
- Careers CTA clicked
- Portal Login clicked
- Qualified lead source / campaign when available

Launch principle

The goal is not only a better-looking website. The new site should be easier to maintain, faster, measurable, mobile-friendly, search-friendly, and ready to hand off cleanly to future developers.

Portal Foundation After Website Launch

Once the public website is stable, begin the internal operations portal as a separate application. The portal should become the source of truth for business operations instead of relying on multiple Excel files.

Core portal workflow

CLIENT
EMPLOYEE / CONSULTANT + ASSIGNMENT
TIMESHEET
APPROVAL
INVOICE
PAYMENT
REPORTING

Portal modules

Business operations	Platform capabilities
• Dashboard • Clients • Employees / Consultants • Assignments • Timesheets • Approvals • Invoices • Payments	• Documents • Reports • Users • Roles & permissions • Audit log • Excel import / export • Email notifications • Integrations

Portal storage direction

Use Google Drive as the initial document provider if that is operationally convenient, but store document metadata and the external file identifier in PostgreSQL. Keep storage behind an application abstraction so the team can migrate to Amazon S3 later without redesigning the business workflow.

Source of truth

Portal = business data and workflows. PostgreSQL = structured business data. Google Drive initially = documents. Gmail = communication. Excel = import/export rather than the primary system.

Definition of Done & Immediate Team Actions

Phase 0 is complete when

- The new website is live at pixelitcenter.com over HTTPS
- All approved pages work on desktop, tablet, and mobile
- Navigation, forms, phone/email links, and Portal Login work correctly
- Content has been reviewed for accuracy, tone, grammar, and consistency
- SEO metadata, sitemap, analytics, and redirects are in place
- Performance and accessibility checks meet the team's agreed baseline
- The old site is backed up and rollback is possible during the launch window
- The repository, environments, deployment process, and ownership are documented

Immediate actions for the team

Phase	Focus	Main Output	Gate
1	Assign owners	Business owner, design owner, engineering owner	This week
2	Inventory current assets	Logo, images, client logos, content, testimonials	Before design
3	Approve sitemap	Home, Services, Staffing, About, Careers, Contact	Before build
4	Create repository + staging	Next.js / TypeScript baseline	Before UI work
5	Create design system	Typography, colors, components	Before pages
6	Build + review homepage	Establish visual direction	First approval
7	Complete remaining pages	Content + responsive QA	Feature complete
8	Launch	SEO, redirects, monitoring, cutover	Production

Recommended first engineering milestone

Milestone 1

A deployed staging.pixelitcenter.com with the new header/navigation, homepage hero, services section, staffing section, contact CTA, responsive layout, and the approved Pixel IT Center design system. This becomes the foundation for all remaining public pages and the future portal UI.

This document is intended to be the team-level implementation direction. Detailed tickets, designs, content copy, infrastructure modules, and portal schemas can be created from this plan as the next layer of execution.